

Unit 1.1

Numbers and their Operations

Paper 1: Basic

1. Calculate $\sqrt[5]{12.5^2 - \frac{6.8}{0.037}}$. [1]
[N22/I/1]
2. Simplify.
 $3y^5 \times 5y^3$ [1]
[N22/I/2(a)]
3. Simplify $\left(\frac{64x^6}{y^3}\right)^{\frac{2}{3}}$. [2]
[N21/I/17]
4. Calculate $\frac{-(-11) - \sqrt{(-11)^2 - 4 \times 16 \times (-50)}}{2 \times 16}$. [1]
[N20/I/1]
5. Write 4^5 as a power of 2. [1]
[N20/I/5(a)]
6. The number of people in a cinema is given as 400, correct to the nearest hundred.
Write down the minimum number of people that could be in the cinema at this time. [1]
[N19/I/1]
7. $< \quad = \quad >$
Choose a symbol from the list above to make a correct statement. [1]
 $33\% \dots\dots\dots \frac{1}{3}$ [N19/I/9(a)]
8. Write the following numbers in order of size, starting with the **smallest**. [1]
 $0.65, \quad \frac{3}{4}, \quad \left(\frac{3}{4}\right)^2, \quad \frac{3}{5}$ [N18/I/1]
9. Simplify $\left(\frac{x^8}{y^6}\right)^{-\frac{3}{2}}$. [2]
[N18/I/2]
10. Given that $\frac{1}{81} = 3^k$, find k . [1]
[N17/I/1]

Paper 1: Intermediate

11. A box measures 84 cm by 60 cm by 36 cm.
The box is completely filled with identical cubes.
Find the minimum number of cubes required. [2]
[N22/I/8]

12. Given that $25^{2x} = 125^7$, find x .

[2]

[N22/I/23]

13. Some bacteria were introduced into a culture.

The number, N , of bacteria t hours after being introduced is given by $N = m \times 2^{3t}$, where m is the number of bacteria introduced.

(a) After 1 hour, the number of bacteria has increased to 2000.

Find m .

[1]

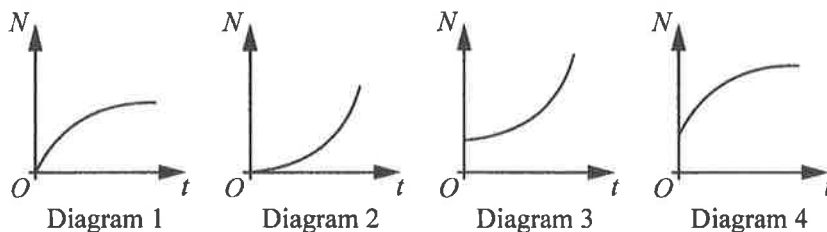
(b) Find, in terms of k , the number of bacteria when $8^t = k$.

[1]

(c) Find the increase in the number of bacteria after 2 hours as a percentage of the number of bacteria originally introduced.

[2]

(d) Which of these diagrams represents the graph of N against t ?



[1]

[N22/I/25]

14. (a) Written as a product of its prime factors, $12 = 2^2 \times 3$.

(i) Write 144 as a product of its prime factors.

[1]

(ii) The highest common factor (HCF) of two numbers is 12.

The lowest common multiple (LCM) of the two numbers is 144.

Both numbers are greater than 20.

Find the two numbers.

[2]

(b) Written as a product of its prime factors, $784 = 2^4 \times 7^2$.

The number $784 \div \frac{p}{q}$ is a perfect cube where p and q are prime numbers.

Find the value of p and the value of q .

[2]

[N21/I/20]

15. (a) Find the prime factors of 1188, giving your answer in index form.

[2]

(b) Two integers, A and B , can be written as products of prime factors.

$$A = p \times q^{r+2} \times 11 \quad B = p^2 \times q^r \times 11$$

The lowest common multiple (LCM) of A and B is 1188.

(i) Write down the values of p , q and r .

[2]

(ii) Find the highest common factor (HCF) of A and B .

[1]

[N20/I/20]

16. (a) Simplify $(x^6)^{\frac{2}{3}}$.

[1]

(b) $3^a = 3^7 + 3^7 + 3^7$

Find the value of a .

[1]

[N19/I/3]

17. (a) Express 360 as a product of its prime factors. [1]
- (b) The number $\frac{360p}{q}$ is a perfect cube.
 p and q are prime numbers.
 Find the value of p and the value of q . [1]
 [N19/I/4]
18. (a) Use prime factors to explain why 28×63 is a perfect square. [3]
- (b) The number $28k$ is a perfect cube.
 Find the smallest positive integer value of k . [2]
 [N17/I/18]
19. (a) Express 784 as a product of its prime factors. [2]
- (b) Using your answer to **part (a)**, explain why 784 is a perfect square. [1]
- (c) m and n are both prime numbers.
 Find the values of m and n so that $784 \times \frac{m}{n}$ is a perfect cube. [1]
 [N16/I/18]
20. A field has an area of 1400 m^2 .
 An average of 60 dandelion plants grow on each square metre.
 Each plant has an average of 20 flowers.
 Joel estimates that these plants will produce 2×10^8 seeds in total.
 Calculate an estimate of the average number of seeds produced by each flower. [2]
 [N15/I/4]

Paper 1: Advanced

21. (a) Express 126 as the product of its prime factors. [1]
- (b) The number $126k$ is a perfect square.
 Find the smallest positive integer value of k . [1]
- (c) x is a number between 200 and 300.
 The highest common factor of x and 126 is 21.
 Find the smallest possible value of x . [2]
 [N18/I/20]
22. Liu has 144 one-centimetre cubes.
 He arranges all of the cubes into a cuboid.
 The perimeter of the top of the cuboid is 22 cm.
 Each side of the cuboid has a length greater than 2 cm.
 Find the height of the cuboid. [2]
 [N16/I/5]
23. (a) Express 450 as the product of its prime factors. [1]
- (b) Find two numbers, both smaller than 100, that have a lowest common multiple of 450 and a highest common factor of 15. [2]
 [N15/I/14]

Paper 2: Basic

1. Simplify $\frac{6a^3b}{5} \div \frac{3a^2}{10b}$. [1]
[N22/II/1(b)]
2. Simplify $\left(\frac{a^9}{27b^{15}}\right)^{-\frac{1}{3}}$. [2]
[N20/II/1(d)]
3. Simplify $\left(\frac{16t^8}{v^{12}}\right)^{-\frac{1}{4}}$. [2]
[N17/II/1(d)]